AI and Renewable Energy across U.S. States, 2015-2025

Results report, as of 2026-11-07. ECON 499R, Emory University. Shunji Lewandowski and Alexander Hendricks. Advisor: Prof. Fohlin.

Summary

We tested two papers on the same state panel. **Once timing is taken into account, data-center job growth isn't associated with electricity use, efficiency or the renewable share across states.** Neither Zhang et al.'s (2026) "blessing or peril" channels nor Fang et al.'s (2025) positive association holds up in our design. All results are associations.

Test	Main number	Holds up?
H1: electricity use, efficiency	-0.0016 (p = 0.795); 0.0022 (p = 0.707)	No association under clustered SEs
H1: renewable share	-0.0101 (p = 0.100)	No. Next year's AI carries it: lead -0.0210 (p = 0.007), lag 0.0015 (p = 0.827)
H1: new renewable capacity	70.8 MW (p = 0.033)	No. 1.00 (p = 0.616) per 1,000 MW of 2015 capacity; its lead is positive too
H2: mediation	Tech-share first stage -0.000157 (p = 0.454)	No channel. Sobel p from 0.322 to 0.991
H3: spillovers	-0.161 (p = 0.035), full sample	Not supported: can't be separated from reverse timing
Fang replication	0.0531 published; -0.0188 (p = 0.317) on current SEDS; 0.0348 (p = 0.057) on SEDS as published in June 2023	Largely reproduces on the June 2023 release (same sign; about two-thirds of Fang's size in col. 6), not on current SEDS (eia.gov/state/seds/seds-change); gone with a linear trend
Fang, our design	Postings -0.0479 (p = 0.109); data-center jobs -0.0567 (p = 0.208)	Neither significant. Data-center jobs are the more reliable measure
Extensions: by type, other outcomes, index	One by-type row with q < 0.10 (postings, wind generation, -0.0477); no row among 30 other outcomes or the index	No. The wind row is exploratory: wrong sign, fails with state trends and with jobs
National trend	AI on renewable share: -0.012 / -0.011 / -0.010 (state FE / drivers / two-way)	The national rise isn't AI; measured drivers track it (consistent with, not proof)
Shift-share IV	KP F = 2.81	Uninformative: F is below 10
Georgia solar, 2035	$37.06/MWh; range $14.51 to $50.40	Arithmetic on cited IEA paths, not a forecast of ours

What we can rule out

Per one within-state SD of AI (0.5212: winsorized lagged AI after removing state and year effects), the 95% CIs are -0.71 to 0.54 percent for electricity use per person, -0.50 to 0.73 percent for GDP per MWh, and -1.16 to 0.11 percentage points for the renewable share.

Why the timing test matters

A standard panel shows a negative association between data-center jobs and the renewable share, and a positive one with new capacity. But next year's AI predicts this year's outcome as well or better. Apart from anticipation, jobs that don't exist yet can't explain what's happening today, so these associations are likely about where and when data-center jobs grow.

What this doesn't cover

Jobs aren't MW, and our sample ends in 2025. Georgia Power reports 15,600 MW of large-load commitments through the mid-2030s as of June 2026, up from 3,600 MW at its October 2023 filing (Q2 2026 Large Load Economic Development Report, filed Aug. 17, 2026, in Georgia PSC Docket 55378, page headers Docket 56002, p. 4 [PDF p. 8]). That's the utility's own reporting, and most of that load is scheduled after our sample ends. Our results describe the past, not that build-out.

1. The question and the data

Is growth in AI-related activity associated with more or less electricity use, energy efficiency and renewable energy? Zhang et al. (2026) argue AI works through tech jobs and density and spills across borders. Fang, Lee and Li (2025) find that AI job postings go with more renewable consumption across states in 2018-2021. We test both, then look at Georgia, a large data-center market with a regulated utility.

Panel: 50 states and DC, 2015-2025, 561 rows. All sources are public: EIA (generation, capacity, sales, SEDS), BLS QCEW (jobs), BEA (GDP, population), Census (land, borders). The Zhang models use the 50 states (N = 550). The workbook's Validation tab has 223 checks: 178 green, 43 yellow, 1 red and 1 not run. The red is the team's NSF file with no state rows, which isn't used. The 20 model variables rate 17 green, 3 yellow and 0 red. Cross-source checks are green, except one reported gap: no net-producer rule reaches Fang's count. The master workbook has every check, source and limitation.

Two AI measures: *private data-center jobs per 1,000 workers* (NAICS 518210), one consistent series, but a proxy for data-center activity rather than AI use; and *AI job postings* (Stanford AI Index), Fang's measure, where almost every year comes from a different report edition. **Data-center jobs are the more reliable measure when we extend the sample.**

Table 1. Main variables, before winsorizing. 50 states.

Variable	Mean	SD	Min	Max	N	Source
Private data-center jobs per 1,000 workers, t-1	2.102	1.246	0.215	7.452	550	BLS QCEW
ln(retail MWh per person)	2.495	0.344	1.802	3.634	550	EIA, BEA
ln(real GDP / retail MWh)	8.484	0.443	7.676	9.555	550	BEA, EIA
Renewable share of generation	0.242	0.239	0.014	0.999	550	EIA
Change in renewable capacity, MW	430.629	934.298	-242.3	9956.0	550	EIA
ln(real GDP per person)	10.979	0.188	10.509	11.457	550	BEA
Manufacturing share of GDP	0.105	0.047	0.016	0.264	550	BEA
Electricity price, cents per kWh	12.047	4.758	7.401	39.72	550	EIA
Population growth	0.006	0.006	-0.014	0.03	550	BEA

2. Methods, in plain English

H1, two-way fixed effects. $Y_{st} = \alpha_s + \gamma_t + \beta\ AI_{s,t-1} + controls + e_{st}$
Compares each state with itself over time, after removing anything that hit all states in a year. β says how much the outcome differs when a state has more data-center jobs the year before. Controls: log GDP per person, manufacturing share, electricity price, population growth. SEs are clustered by state; Driscoll-Kraay is a sensitivity check only, since there are just 11 years. AI and outcomes are winsorized at the 1st and 99th percentiles.

Timing test. Replace AI_{t-1} with AI_{t+1}, then put both in one model. Apart from anticipation, future jobs can't produce today's outcome, so if the lead carries the association, it reflects where and when jobs grow (reverse timing or selection).

H2, mediation. $M = a\ AI + \ldots$ and $Y = b\ M + c\ AI + \ldots$, indirect path = a × b. Does AI go with the mediator, and does the mediator go with the outcome? Mediators: tech-job share (main), HMA (the memo's version, positive by construction) and economic density. Sobel tests and a cluster bootstrap.

H3, spatial Durbin. $Y_{st} = \rho\ W\ Y_{st} + \beta\ AI_{s,t-1} + \theta\ W\ AI_{s,t-1} + controls + FE + e_{st}$
W averages over neighbors, weighted by inverse distance (main) or shared border (check). Because outcomes feed back through ρ, we report direct (own state), indirect (neighbors, the spillover) and total. Plug-in estimates with simulated CIs.

Fang test. $\ln RE_{st} = \alpha_s + \beta\ \ln AI_{st} + \varphi\ (\ln AI_{st} \times EPU_{st}) + controls + e_{st}$
RE is SEDS renewable consumption. State effects only and conventional SEs, as in Fang. The bridge changes one thing at a time: year effects, lag, clustering, longer sample, no DC. That shows which step changes the answer.

Shift-share IV. Instrument = a state's 2015 share of national data-center jobs × national jobs outside that state. It's useful only if it predicts AI: a KP F below 10 means uninformative.

Georgia learning curve. $C_t = C_{base}\ (X_t\ /\ X_{base})^b$, $b = \ln(1 - LR)\ /\ \ln 2$
Cost falls by the learning rate LR each time global capacity X doubles. X follows IEA's scenario paths. It's arithmetic: it has no AI variable, and learning curves are correlations.

3. Results: the Zhang test

H1. Electricity use and efficiency show no association under clustered SEs. The renewable share is negative but not significant (p = 0.100; p = 0.001 under Driscoll-Kraay, sensitivity only). New capacity is positive in raw MW, driven by large-grid states, and not robust to scaling. No AI-squared curve passes the Lind-Mehlum test.

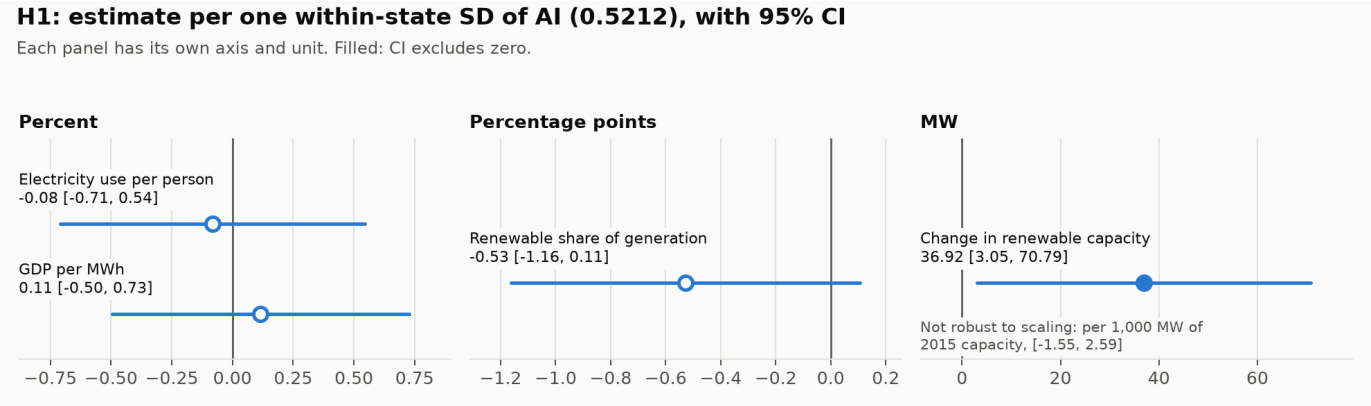


Figure 1. H1: AI(t-1) per one within-state SD of AI (0.5212: winsorized lagged AI after removing state and year effects), with 95% CIs. One panel per unit. Two-way FE, ctrl4, clustered by state.

Table 2. H1: AI(t-1), two-way FE, four controls, clustered by state. Coefficient (SE). * p<0.1, ** p<0.05, *** p<0.01.

Outcome	Group	AI	N
ln_ec	Core	-0.0016 (0.0060)	550
ln_ee	Core	0.0022 (0.0058)	550
renew_share	Core	-0.0101 (0.0060)	550
new_renew_mw	Extra	70.84** (32.34)	550
new_renew_mw per 1,000 MW of 2015 capacity	Extra	0.9962 (1.9762)	550
new_renew_mw per million people	Extra	10.87 (8.60)	550

H2. No channel is supported. AI isn't associated with the tech-job share or density, and HMA's first stage is positive by construction. All bootstrap CIs include zero.

Table 3. H2: mediation. Coefficient (SE), Sobel p and bootstrap CI.

Mediator	Outcome	AI to mediator	Mediator to outcome	Indirect	Sobel p	Bootstrap 95% CI
tech_share (H2a main)	ln_ec	-0.00016 (0.00021)	-3.872 (3.578)	0.00061	0.536	[-0.0020, 0.0032]
tech_share (H2a main)	ln_ee	-0.00016 (0.00021)	6.006 (4.057)	-0.00094	0.501	[-0.0044, 0.0025]
tech_share (H2a main)	renew_share	-0.00016 (0.00021)	-1.179 (5.025)	0.00019	0.823	[-0.0019, 0.0044]
HMA (H2a, memo version)	ln_ec	0.04164*** (0.00322)	-0.001 (0.128)	-0.00006	0.991	[-0.0104, 0.0124]
HMA (H2a, memo version)	ln_ee	0.04164*** (0.00322)	0.124 (0.194)	0.00518	0.522	[-0.0104, 0.0219]
HMA (H2a, memo version)	renew_share	0.04164*** (0.00322)	-0.135 (0.163)	-0.00561	0.409	[-0.0217, 0.0066]
ln_ea (H2b)	ln_ec	-0.00336 (0.00318)	-0.333** (0.136)	0.00112	0.332	[-0.0008, 0.0051]
ln_ea (H2b)	ln_ee	-0.00336 (0.00318)	0.403*** (0.142)	-0.00136	0.322	[-0.0057, 0.0010]
ln_ea (H2b)	renew_share	-0.00336 (0.00318)	-0.254 (0.214)	0.00085	0.430	[-0.0009, 0.0050]

H3. Renewable share is clustered across states (Moran's I from 0.094 to 0.157). On 2015-2023, the only years the timing test can run, the renew_share spillover isn't significant, and with border weights neighbors' future AI carries the association rather than their past AI. The full-sample estimate (-0.161, p = 0.035) can't be separated from this reverse timing. H3 is not supported. On inverse-distance weights, the test can't tell lag from lead: both CIs are wide. For ln_ec and ln_ee, the main-spec spillover CIs include zero. **Exploratory note:** with lead and lag in one model, the SLX shows neighbors' past AI as significant for ln_ec (0.0419, p = 0.034 inverse distance; 0.0234, p = 0.024 border) and for border-weights ln_ee (-0.0209, p = 0.033). The SDM doesn't confirm it: those indirect CIs include zero. With this many tests, a few p < 0.05 are expected by chance, so we don't read these as spillovers.

Table 4. H3: spatial Durbin, full sample. Estimate [95% CI].

W	Outcome	rho	Direct	Indirect	Total	p, indirect
Inverse distance, 50 states (main)	ln_ec	0.137	-0.0026 [-0.010, 0.006]	0.0178 [-0.147, 0.226]	0.0152 [-0.149, 0.227]	0.963
Inverse distance, 50 states (main)	ln_ee	-0.018	0.0037 [-0.005, 0.012]	-0.0064 [-0.229, 0.153]	-0.0027 [-0.236, 0.161]	0.997
Inverse distance, 50 states (main)	renew_share	0.617	-0.0155 [-0.024, -0.007]	-0.1613 [-0.344, -0.035]	-0.1768 [-0.363, -0.046]	0.035
Border, 48 states (check)	ln_ec	0.302	-0.0027 [-0.014, 0.027]	-0.0022 [-0.037, 0.190]	-0.0048 [-0.049, 0.212]	1.000
Border, 48 states (check)	ln_ee	0.234	0.0032 [-0.007, 0.016]	0.0032 [-0.024, 0.050]	0.0064 [-0.030, 0.065]	0.867
Border, 48 states (check)	renew_share	0.453	-0.0137 [-0.027, 0.001]	-0.0448 [-0.091, 0.010]	-0.0585 [-0.116, 0.011]	0.087

Table 5. Timing test for the renew_share spillover, 2015-2023 (N = 450 inverse distance; border weights use 48 states). SLX: coefficient (clustered SE). SDM: indirect estimate [95% CI].

Model and term	Inverse distance	Border
SLX: W*AI(t-1) alone	-0.0130 (0.0331)	-0.0138 (0.0177)
SLX: W*AI(t+1) swap	-0.0196 (0.0353)	-0.0353* (0.0176)
SLX both: lag	-0.0105 (0.0343)	0.0098 (0.0177)
SLX both: lead	-0.0063 (0.0379)	-0.0406** (0.0178)
SDM: lag alone	-0.082 [-0.588, 0.447]	-0.028 [-0.115, 0.072]
SDM: swap	-0.091 [-0.554, 0.489]	-0.053 [-0.108, 0.042]
SDM both: lag	-0.080 [-0.721, 0.648]	-0.001 [-0.070, 0.083]
SDM both: lead	-0.009 [-0.431, 0.374]	-0.043 [-0.079, -0.002]

Robustness and placebo. The RPS control (the LBNL current RPS/CES schedule), all-ownership AI and holding Gavins Point in Nebraska change no result. For renew_share, next year's AI is -0.0197 (p = 0.006); in one model, the lead is -0.0210 (p = 0.007) and the lag 0.0015 (p = 0.827). For new_renew_mw, the lead is 86.8 (p = 0.028); in one model, 55.9 (p = 0.219) and 37.5 (p = 0.191): neither significant.

Weather check. With NOAA heating and cooling degree days on the 48 contiguous states, the ln_ec coefficient is -0.0021 (p = 0.711), against -0.0009 (p = 0.880) on the same states without them. Both degree-day terms are significant (p < 0.001), but the AI coefficient doesn't change.

Table 6. Robustness: AI coefficient (SE) by specification. * p<0.1, ** p<0.05, *** p<0.01. Last row: the placebo.

Specification	ln_ec	ln_ee	renew_share	new_renew_mw	per 1,000 MW
Baseline	-0.0016 (0.0060)	0.0022 (0.0058)	-0.0101 (0.0060)	70.84** (32.34)	1.00 (1.98)
Drop 2020	-0.0020 (0.0061)	0.0024 (0.0059)	-0.0108* (0.0062)	71.14** (33.28)	1.07 (1.88)
Drop AK, HI	-0.0009 (0.0058)	0.0015 (0.0057)	-0.0097 (0.0058)	68.80** (30.86)	1.02 (1.93)
Drop VA	-0.0037 (0.0056)	0.0042 (0.0055)	-0.0101 (0.0061)	67.77** (32.01)	0.85 (1.98)
Border SLX, own AI	-0.0017 (0.0059)	0.0025 (0.0055)	-0.0135** (0.0058)	71.47* (37.82)	0.39 (2.07)
Long difference (fixed)	-0.0079 (0.0077)	0.0068 (0.0078)	-0.0137 (0.0084)	–	–
RPS: LBNL current RPS/CES schedule	-0.0021 (0.0058)	0.0027 (0.0057)	-0.0103* (0.0061)	73.78** (33.14)	1.25 (2.02)
RPS: June 2023 edition	-0.0021 (0.0058)	0.0028 (0.0057)	-0.0102 (0.0061)	73.99** (33.18)	1.27 (2.02)
RPS: current schedule, no NY	-0.0021 (0.0058)	0.0028 (0.0057)	-0.0103* (0.0061)	71.92** (33.05)	1.23 (2.03)
AI: all ownerships	-0.0012 (0.0060)	0.0016 (0.0058)	-0.0100 (0.0060)	70.79** (32.16)	–
Gavins Point held in NE	–	–	–	71.02** (32.41)	–
Driscoll-Kraay SE	-0.0016 (0.0013)	0.0022** (0.0009)	-0.0101*** (0.0023)	70.84** (30.56)	1.00 (1.37)
Conley 500 km SE	-0.0016 (0.0064)	0.0022 (0.0062)	-0.0101* (0.0061)	70.84*** (26.34)	1.00 (2.02)
Placebo: AI(t+1)	-0.0028 (0.0065)	0.0040 (0.0067)	-0.0197*** (0.0069)	86.81** (38.42)	0.38 (1.79)

4. Results: the Fang test

Replication. Fang's positive coefficient doesn't reproduce on current SEDS. In October 2023, starting with the 2022 data cycle and for all years, EIA moved wind, solar, hydro and geothermal from fossil fuel equivalency to captured energy at 3,412 Btu/kWh (EIA notice: eia.gov/state/seds/seds-change). **SEDS as published in June 2023, the archived release and our main old-basis series, largely reproduces Fang:** the same sign, a similar size in columns 1-2 (0.0346 and 0.0435), and about two-thirds of Fang's size in column 6 (0.0348, p = 0.057). Our year-by-year rebuild of the old accounting on today's data gives 0.0137 (p = 0.500): 61% of the column-6 gap comes from the accounting change and the rest from EIA's later data revisions. k = 2.3-2.9, our first approximation, is kept as a check. Fang's paper says EIA's renewable data ended in 2021 (PDF p. 3), and its Table 2 constants imply a mean log outcome of 11.79, which is consistent with the old basis (11.78-11.90), not the current one (11.44), assuming the usual fixed-effects constant. But Fang's spec has no year effects: on the June 2023 basis a linear trend turns column 6 to -0.0419 (p = 0.121), and pet-care employment is associated with renewables more strongly than AI (0.2980, p = 0.003).

Fang's supplement. Against Fang's Table A1, our ln AI postings, EP, POP and EDU match, GDP differs by the 2012-vs-2017 dollar base, and ln EPU is 0.708 lower, close to ln 2 (consistent with a doubled index, which state FE in logs absorb). Supplement B1's coal and gas coefficients fall inside our CIs; petroleum doesn't.

Splits, negative indicator, weather, event study. On current SEDS none of Fang's subgroup splits reproduces. On the June 2023 basis (added after seeing Table 2, so exploratory), low-consumption, low-R&D, high-income and Midwest states match Fang's side, but **Fang's headline splits, electricity price and net producer, don't reproduce**, nor does the extreme-weather split (NOAA Storm Events, our counting rule). Fossil-fuel consumption is never negative and significant. In the event study on 2019 postings exposure, the pre-2021 coefficients are jointly significant on current SEDS (p = 0.014), so the post-2021 gap looks like a trend already under way.

Table 7. Fang's Table 2 and our replication: AI coefficient (SE), state FE, conventional SEs. k is our approximation of the old SEDS accounting. * p<0.1, ** p<0.05, *** p<0.01.

Version	Measure, sample	N	(1) AI	(2) + EPU	(3) + EP	(4) + GDP	(5) + POP	(6) + EDU
Fang Table 2 (published)	AI postings, 2018-2021	204	0.0428*** (0.015)	0.0452*** (0.015)	0.0396** (0.015)	0.0548*** (0.016)	0.0561*** (0.019)	0.0531*** (0.019)
Current SEDS	AI postings, 2018-2021	204	-0.0288** (0.0141)	-0.0100 (0.0139)	-0.0189 (0.0159)	-0.0159 (0.0183)	-0.0183 (0.0186)	-0.0188 (0.0187)
Current SEDS, same vintage	AI postings, 2019-2021	153	0.0023 (0.0266)	-0.0240 (0.0246)	-0.0762* (0.0400)	-0.0680* (0.0401)	-0.0851* (0.0435)	-0.0852* (0.0437)
Current SEDS, as first published	AI postings, 2018-2021	204	-0.0355*** (0.0107)	-0.0152 (0.0116)	-0.0357** (0.0152)	-0.0345** (0.0158)	-0.0412** (0.0166)	-0.0414** (0.0166)
Approx. old SEDS, k = 2.3	AI postings, 2018-2021	204	0.0180 (0.0144)	0.0284* (0.0149)	0.0166 (0.0169)	0.0188 (0.0195)	0.0162 (0.0198)	0.0165 (0.0199)
Approx. old SEDS, k = 2.6	AI postings, 2018-2021	204	0.0251* (0.0145)	0.0342** (0.0151)	0.0219 (0.0171)	0.0237 (0.0198)	0.0213 (0.0201)	0.0216 (0.0202)
Approx. old SEDS, k = 2.9	AI postings, 2018-2021	204	0.0313** (0.0147)	0.0393** (0.0153)	0.0265 (0.0174)	0.0281 (0.0201)	0.0257 (0.0204)	0.0262 (0.0205)
Current SEDS	Data-center jobs, 2018-2021	204	-0.0950* (0.0508)	-0.0503 (0.0487)	-0.0735 (0.0526)	-0.0674 (0.0536)	-0.0651 (0.0541)	-0.0655 (0.0542)
Approx. old SEDS, k = 2.6	Data-center jobs, 2018-2021	204	0.0144 (0.0528)	0.0299 (0.0537)	-0.0216 (0.0572)	-0.0272 (0.0583)	-0.0220 (0.0587)	-0.0218 (0.0589)

Bridge. Neither measure is significant in the final spec. **Data-center jobs are the more reliable measure:** the postings extension mixes real change with edition changes. The share-of-national column isn't an independent test: with year effects, it equals the postings column by construction.

Table 8. Bridge from Fang's spec to ours, one change at a time. Coefficient (SE) [N]. * p<0.1, ** p<0.05, *** p<0.01.

Step	ln(AI postings)	ln(share of national postings)	ln(data-center jobs)	ln(jobs per 1,000)
Fang's spec, 2018-2021	-0.0188 (0.0187) [204]	-0.0881*** (0.0334) [204]	-0.0655 (0.0542) [204]	-0.0743 (0.0528) [204]
+ year FE	-0.1095*** (0.0371) [204]	-0.1094*** (0.0371) [204]	-0.0628 (0.0558) [204]	-0.0716 (0.0558) [204]
2019-2021 (years the lag can use)	-0.1055** (0.0503) [153]	-0.1054** (0.0503) [153]	-0.0242 (0.0754) [153]	-0.0360 (0.0754) [153]
+ lag AI one year	-0.0761 (0.0474) [153]	-0.0761 (0.0474) [153]	-0.1121 (0.0694) [153]	-0.1215* (0.0696) [153]
+ cluster by state	-0.0761 (0.0518) [153]	-0.0761 (0.0518) [153]	-0.1121 (0.0850) [153]	-0.1215 (0.0886) [153]
+ extend the sample	-0.0496* (0.0290) [306]	-0.0492* (0.0290) [306]	-0.0279 (0.0499) [510]	-0.0301 (0.0509) [510]
+ drop DC	-0.0479 (0.0293) [300]	-0.0475 (0.0293) [300]	-0.0567 (0.0444) [500]	-0.0596 (0.0453) [500]
Check: 2019-2024, with DC	–	–	-0.0361 (0.0570) [306]	-0.0427 (0.0583) [306]
Check: 2019-2024, no DC	–	–	-0.0388 (0.0593) [300]	-0.0463 (0.0606) [300]

EPU moderation, by type and placebo. AI × EPU is positive in Fang's spec, and not significant under two-way FE, for either measure. By type, Fang's spec shows solar positive and biomass negative with postings, partly like Fang's pattern. Under two-way FE only biomass with postings stands out. The placebo isn't informative here: AI(t-1) isn't significant either.

Net-producer split (exploratory). Fang's rule is electricity production above consumption from EIA, series not stated (PDF p. 4), with 76 net-producer state-years (Table 3). Our rule, generation above retail sales on each state's 2018-2021 average, gives 120. SEDS total energy production above consumption gives 40, closer by count but not Fang's concept; SEDS net interstate electricity flow below zero gives 104. None reaches 76. In Fang's spec with postings, the producer-minus-consumer gap is 0.0864 ($p = 0.003$) on our rule, and isn't significant on the other two ($p = 0.316$; $p = 0.177$).

Table 9. AI × EPU (centered), with the AI slope at the 25th and 75th percentiles of EPU. * p<0.1, ** p<0.05, *** p<0.01.

Spec	EPU	AI measure	AI x EPU	AI slope, EPU p25	AI slope, EPU p75	N
Fang spec	Composite	AI postings	0.0215*** (0.0075)	-0.0102 (0.0185)	0.0082 (0.0205)	204
Fang spec	Composite	Data-center jobs	0.0208*** (0.0073)	-0.0761 (0.0531)	-0.0583 (0.0530)	204
Fang spec	National	AI postings	0.0211** (0.0081)	-0.0150 (0.0180)	0.0001 (0.0195)	204
Fang spec	National	Data-center jobs	0.0201** (0.0080)	-0.0800 (0.0531)	-0.0656 (0.0528)	204
Fang spec	State	AI postings	0.0193*** (0.0062)	-0.0093 (0.0186)	0.0150 (0.0215)	204
Fang spec	State	Data-center jobs	0.0199*** (0.0061)	-0.0798 (0.0521)	-0.0549 (0.0523)	204
Two-way FE	Composite	AI postings	0.0154 (0.0100)	-0.0601* (0.0306)	-0.0503* (0.0293)	300
Two-way FE	Composite	Data-center jobs	0.0149 (0.0118)	-0.0642 (0.0465)	-0.0546 (0.0440)	500
Two-way FE	National	AI postings	0.0159 (0.0104)	-0.0589* (0.0307)	-0.0488 (0.0293)	300
Two-way FE	National	Data-center jobs	0.0153 (0.0126)	-0.0652 (0.0466)	-0.0557 (0.0441)	500
Two-way FE	State	AI postings	0.0117 (0.0083)	-0.0619** (0.0305)	-0.0508* (0.0291)	300
Two-way FE	State	Data-center jobs	0.0109 (0.0097)	-0.0614 (0.0461)	-0.0523 (0.0435)	500

Table 10. By renewable type: coefficient (SE) [N]. * p<0.1, ** p<0.05, *** p<0.01.

Type	Fang Table 7	Fang spec, postings	Fang spec, jobs	Two-way FE, postings	Two-way FE, jobs
Wind	0.1542*** [164]	0.0836* (0.0469) [164]	0.0624 (0.1518) [164]	-0.0107 (0.0605) [250]	-0.3406 (0.2295) [412]
Solar	0.4970*** [204]	0.3829*** (0.0712) [204]	0.2188 (0.2250) [204]	0.1193 (0.1438) [300]	-0.0964 (0.2584) [500]
Geothermal	0.0210 [204]	0.0414 (0.0679) [204]	0.2035 (0.1964) [204]	0.1974 (0.1828) [300]	0.0638 (0.0625) [500]
Biomass	-0.0535*** [204]	-0.1135*** (0.0260) [204]	-0.0796 (0.0799) [204]	-0.0785** (0.0313) [300]	-0.0159 (0.0473) [500]
Hydro	0.1160*** [192]	0.0212 (0.0421) [192]	0.1433 (0.1224) [192]	0.0142 (0.0625) [288]	-0.0617 (0.0847) [480]

Table 11. Fang placebo, final spec: coefficient (SE).

AI measure	N	AI(t-1) alone	AI(t+1) alone	Both: lead	Both: lag	Within corr
AI postings	300	-0.0479 (0.0293)	-0.0518 (0.0356)	-0.0353 (0.0305)	-0.0335 (0.0229)	0.708
Data-center jobs	496	-0.0573 (0.0440)	-0.0530 (0.0453)	-0.0146 (0.0512)	-0.0494 (0.0519)	0.723

5. Results: extensions (Nov 7)

Every model was pre-registered in `EXPERIMENTS.md` section 14 before it ran; deviations X1-X5 and F27 are logged there. Main spec: two-way FE, AI(t-1), ctrl4, clustered by state, winsorized; both data-center jobs and postings per 1,000 workers. A finding needs BH q < 0.10 within its family and a clearly smaller lead.

Which renewable? One by-type row passes: postings and ln wind generation (-0.0477, q = 0.086). It is exploratory: it runs against the expected sign, loses significance with state trends (p = 0.135), and the jobs version fails the lead test. Biofuels are null, so the negative SEDS total isn't transport fuel. **Georgia:** its solar share rose 0.076 in 2015-2025; the AI association could account for at most 25% with jobs. Georgia Power's 2025 IRP (Docket 56002) plans for about 8,200 MW of load growth through winter 2030/2031, with RFPs for up to 9,500 MW of capacity that the IRP doesn't split by fuel, 1,323 MW of new combustion turbines, about 780 MW of batteries (p. 128 [PDF p. 135]) and coal extensions.

What AI is associated with. None of 30 rows (sales by sector, prices, generation mix, additions, imports, CO2) has q < 0.10. Per one within-state SD of data-center jobs, commercial sales per person move between -1.6% and 1.2%.

The national trend. The renewable-share coefficient barely moves across state FE only (-0.012), measured drivers (-0.011) and two-way FE (-0.010). Drivers (RPS targets; solar and wind LCOE from LBNL; the ITC and PTC from CRS IF10479 and R43453 and P.L. 116-260 and 117-169; Henry Hub) give a path that correlates with the year effects at 0.95, consistent with their role but not proof, since any rising series tracks a national trend.

Relevance and a broader measure. Pooled, data-center jobs don't predict commercial sales per person (p = 0.764); Virginia's per-state slope ranks 3 of 50 and Texas 7. An "infrastructure + talent" index (2018-2023, 43 states; alpha 0.73; correlation with BTOS AI use 0.55) doesn't predict commercial sales either (p = 0.715), and no index row has q < 0.10, on all 50 states or without CA, CO and DE. Groupings by AI investment and a land moderator find nothing

that survives BH, except an Alaska-driven land result we read as fragile. The distillery placebo stays significant with state trends (p = 0.014), so state-specific trends remain a live problem. **Not run:** the Queued Up outcome (no scripted download), the innovation dimension (no inventor locations).

Table 12. Fang Table 2 AI and control coefficients by SEDS basis, Fang spec. June 2023 is the main old basis; k = 2.6 is a check.

ai_measure	column	fang	current	k26	rebuilt	archived	archived_fang_in_ci
ln AI postings	1	0.0428	-0.0288	0.0251	0.0149	0.0346	True
ln AI postings	2	0.0452	-0.01	0.0342	0.0269	0.0435	True
ln AI postings	3	0.0396	-0.0189	0.0219	0.012	0.0316	True
ln AI postings	4	0.0548	-0.0159	0.0237	0.0155	0.0361	True
ln AI postings	5	0.0561	-0.0183	0.0213	0.0133	0.0338	True
ln AI postings	6	0.0531	-0.0188	0.0216	0.0137	0.0348	True
ln data-center jobs	1	0.0428	-0.095	0.0144	0.0022	0.0224	True
ln data-center jobs	2	0.0452	-0.0503	0.0299	0.0259	0.0355	True
ln data-center jobs	3	0.0396	-0.0735	-0.0216	-0.0272	-0.0209	True
ln data-center jobs	4	0.0548	-0.0674	-0.0272	-0.0285	-0.0269	True
ln data-center jobs	5	0.0561	-0.0651	-0.022	-0.0244	-0.0213	True
ln data-center jobs	6	0.0531	-0.0655	-0.0218	-0.0242	-0.0207	True

Table 13. Gap from current to June 2023 SEDS: accounting change vs later data revisions.

ai_measure	column	current	rebuilt	june2023	share_accounting_rebuild	share_later_data_revisions	june2023_as_share_of_fang
ln AI postings	1	-0.0288	0.0149	0.0346	0.69	0.31	0.808
ln AI postings	2	-0.01	0.0269	0.0435	0.691	0.309	0.962
ln AI postings	6	-0.0188	0.0137	0.0348	0.606	0.394	0.655
ln data-center jobs	1	-0.095	0.0022	0.0224	0.828	0.172	0.524
ln data-center jobs	2	-0.0503	0.0259	0.0355	0.888	0.112	0.786
ln data-center jobs	6	-0.0655	-0.0242	-0.0207	0.922	0.078	-0.389

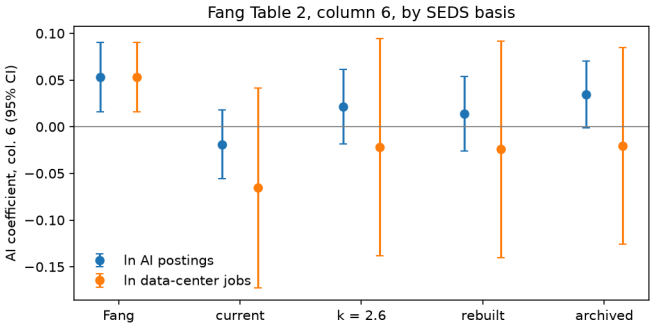


Figure 2. Fang Table 2, column 6, AI coefficient by SEDS basis, with 95% CIs. June 2023 = SEDS as published before EIA's accounting change.

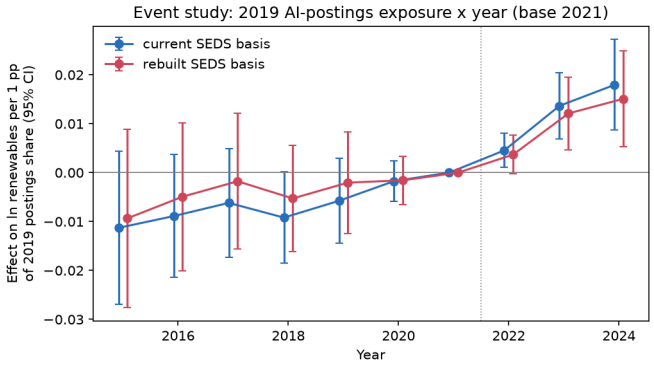


Figure 3. Event study on 2019 postings exposure, base 2021. Pre-2021 coefficients rise toward zero (jointly significant on current SEDS), so post-2021 gains look like a continuing trend.

Table 14. AI coefficient: state FE only, full measured-drivers model, two-way FE.

outcome	ai_measure	drivers	statefe	twfe
ln_gen_solar	ai518210	+0.03032 (p=0.823)	-0.04571 (p=0.794)	+0.04204 (p=0.746)
ln_gen_solar	aipostp1000	-0.1026* (p=0.083)	+0.1229*** (p=0.007)	-0.04603 (p=0.565)
ln_gen_wind	ai518210	-0.1937*** (p=0.010)	-0.2173*** (p=0.008)	-0.1805** (p=0.016)
ln_gen_wind	aipostp1000	-0.04155*** (p=0.004)	-0.001927 (p=0.869)	-0.0477*** (p=0.002)
renew_share	ai518210	-0.0111* (p=0.073)	-0.01242** (p=0.046)	-0.01013 (p=0.100)
renew_share	aipostp1000	-0.004509** (p=0.037)	+0.0005188 (p=0.774)	-0.005681* (p=0.089)

Table 15. Ranks of VA, TX and GA among per-state slopes of commercial sales on AI(t-1) (1 = largest).

sample	ai_measure	state	slope	rank_desc	rank_of
all50	ai518210	VA	0.242025391104506	3	50
all50	ai518210	TX	0.0979533597337702	7	50
all50	ai518210	GA	0.0201522807364743	21	50
all50	aipostp1000	VA	0.0562854487811715	4	50
all50	aipostp1000	TX	0.0129406221856079	10	50
all50	aipostp1000	GA	0.0145098853963786	8	50
all50	indexA	VA	0.362557062267847	1	42
all50	indexA	TX	0.040929054632924	11	42
all50	indexA	GA	0.0675928638507051	6	42
noCACODE	ai518210	VA	0.237482932442187	4	47
noCACODE	ai518210	TX	0.089071721489435	7	47
noCACODE	ai518210	GA	0.0175738711497005	18	47
noCACODE	aipostp1000	VA	0.0549239247284999	4	47
noCACODE	aipostp1000	TX	0.01110092048142	9	47
noCACODE	aipostp1000	GA	0.0127190928761217	8	47
noCACODE	indexA	VA	0.355870367073742	1	40
noCACODE	indexA	TX	0.0347500416409869	11	40
noCACODE	indexA	GA	0.0615866399517439	6	40

Table 16. AI index family: rows, smallest q and rows with q below 0.10, by sample and index.

sample	index	rows	min_q	q_below_10
all50	index_a	35	0.863	0
all50	index_b	35	0.863	0
all50	sub_infra	35	0.863	0
all50	sub_talent	35	0.863	0
noCACODE	index_a	35	0.9	0
noCACODE	index_b	35	0.9	0
noCACODE	sub_infra	35	0.9	0
noCACODE	sub_talent	35	0.9	0

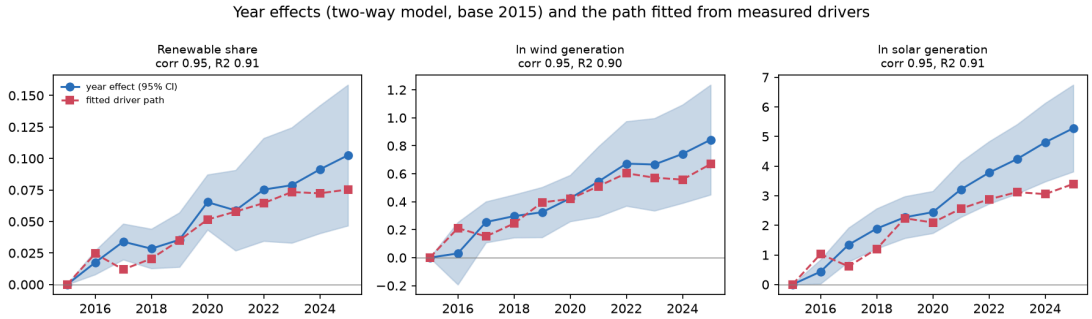


Figure 4. Year effects (two-way model, base 2015) and the path fitted from measured drivers. Correlation is consistent with the drivers' role, not proof of it.

6. Results: stretch goals and Georgia

Shift-share IV. The first stage is too weak (KP F = 2.81), so the IV is uninformative and can't settle the timing question. KP rank LM is computed directly, because the package's version errors with clustering. **Moderators.** No broadband or R&D interaction is significant. The Hansen threshold wasn't run: the team dropped it.

Table 17. Shift-share IV first stage. * p<0.1, ** p<0.05, *** p<0.01.

Base year	First-stage coefficient	KP F	KP rank LM	Rank p	N
2015	0.0116* (0.0069)	2.81	3.05	0.081	550
2014 (check)	0.0100 (0.0066)	2.28	2.56	0.11	550

Table 18. OLS, reduced form and IV, coefficient (SE). * p<0.1, ** p<0.05, *** p<0.01. The IV column is uninformative.

Outcome	OLS	Reduced form	IV (uninformative)
ln_ec	-0.0016 (0.0060)	0.0005 (0.0007)	0.0392 (0.0735)
ln_ee	0.0022 (0.0058)	-0.0006 (0.0007)	-0.0504 (0.0734)
renew_share	-0.0101 (0.0060)	0.0013* (0.0007)	0.1115 (0.0930)
new_renew_mw	70.84** (32.34)	16.58*** (3.52)	1425.18* (783.08)

Table 19. Broadband and R&D moderation, each with its main effect. * p<0.1, ** p<0.05, *** p<0.01.

Moderator	Outcome	AI x moderator	Moderator main effect	AI slope at p25	AI slope at p75	N
Broadband, 2016-2024, no 2020	ln_ec	-0.0002 (0.0009)	0.0027 (0.0026)	-0.0010 (0.0053)	-0.0022 (0.0070)	400
Broadband, 2016-2024, no 2020	ln_ee	0.0003 (0.0009)	-0.0015 (0.0033)	0.0007 (0.0047)	0.0031 (0.0073)	400
Broadband, 2016-2024, no 2020	renew_share	0.0008 (0.0012)	0.0018 (0.0031)	-0.0146* (0.0077)	-0.0082 (0.0086)	400
Broadband, 2016-2024, no 2020	new_renew_mw	9.16 (5.63)	37.16* (20.79)	45.15 (41.76)	115.89*** (38.96)	400
R&D share of GDP, 2017-2023	ln_ec	0.0006 (0.0022)	-0.0096* (0.0053)	0.0001 (0.0061)	0.0014 (0.0056)	350
R&D share of GDP, 2017-2023	ln_ee	-0.0015 (0.0025)	0.0084 (0.0061)	0.0023 (0.0060)	-0.0007 (0.0055)	350
R&D share of GDP, 2017-2023	renew_share	0.0002 (0.0024)	0.0020 (0.0081)	-0.0100 (0.0070)	-0.0096 (0.0066)	350
R&D share of GDP, 2017-2023	new_renew_mw	-13.62 (19.03)	22.03 (49.86)	124.91*** (43.96)	98.45** (41.07)	350

Georgia solar. The base is Georgia's capacity-weighted LCOE without tax credits, $63.81/MWh, for 17 projects in service in 2023-2025 (LBNL). Capacity paths use IEA's *World Energy Outlook 2025*, Annex A, solar PV capacity: Low is Current Policies (Table A.3a, p. 417), Central is Stated Policies (Table A.3b, p. 423), High is Net Zero Emissions by 2050 (Table A.3c, p. 429). **These aren't forecasts of ours.** National LCOE rose from $49.0 in 2022 to $61.0 in 2025 while global capacity more than doubled, so the curve assumes costs resume falling. Georgia Power's 2025 IRP plans RFPs for up to 4,000 MW of renewable resources by 2035, which is the utility's own plan.

Table 20. Georgia solar LCOE, 2025 $/MWh, by learning rate and IEA path. Not forecasts of ours. Paths: IEA WEO 2025, Tables A.3a, A.3b and A.3c.

Learning rate	IEA path	2025	2030	2035	Change 2025-2035, %
12%	Low	61.44	55.64	50.40	-18.0
12%	Central	61.44	55.17	49.54	-19.4
12%	High	61.44	53.44	46.49	-24.3
19% (CapEx)	Low	59.95	50.92	43.25	-27.9
19% (CapEx)	Central	59.95	50.20	42.04	-29.9
19% (CapEx)	High	59.95	47.64	37.86	-36.8
24% (main)	Low	58.82	47.56	38.45	-34.6
24% (main)	Central	58.82	46.69	37.06	-37.0
24% (main)	High	58.82	43.61	32.33	-45.0
45% (2014-2020 epoch)	Low	53.45	33.64	21.17	-60.4
45% (2014-2020 epoch)	Central	53.45	32.31	19.53	-63.5
45% (2014-2020 epoch)	High	53.45	27.85	14.51	-72.9

7. Key figures

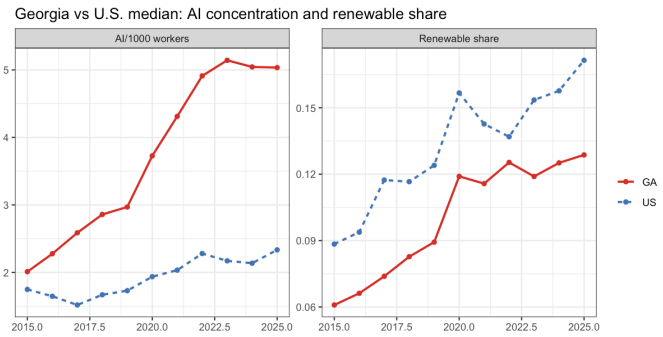


Figure 5. Georgia and the U.S.: private data-center jobs per 1,000 workers and renewable share.

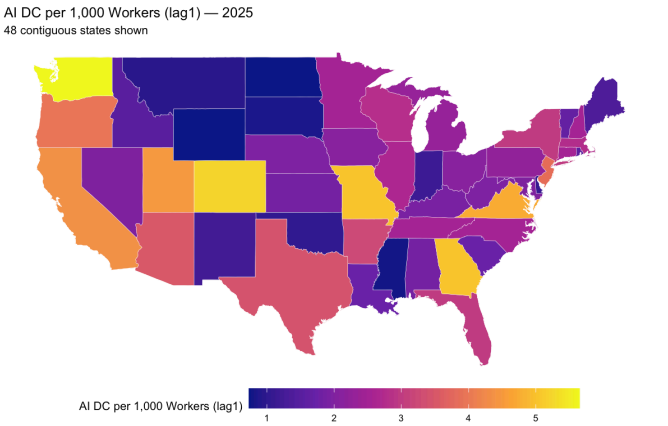


Figure 6. Private data-center jobs per 1,000 workers, lagged, 2025 (BLS QCEW). AK, ND, SD and WY are suppressed only in 2025 itself, so this lag uses 2024.

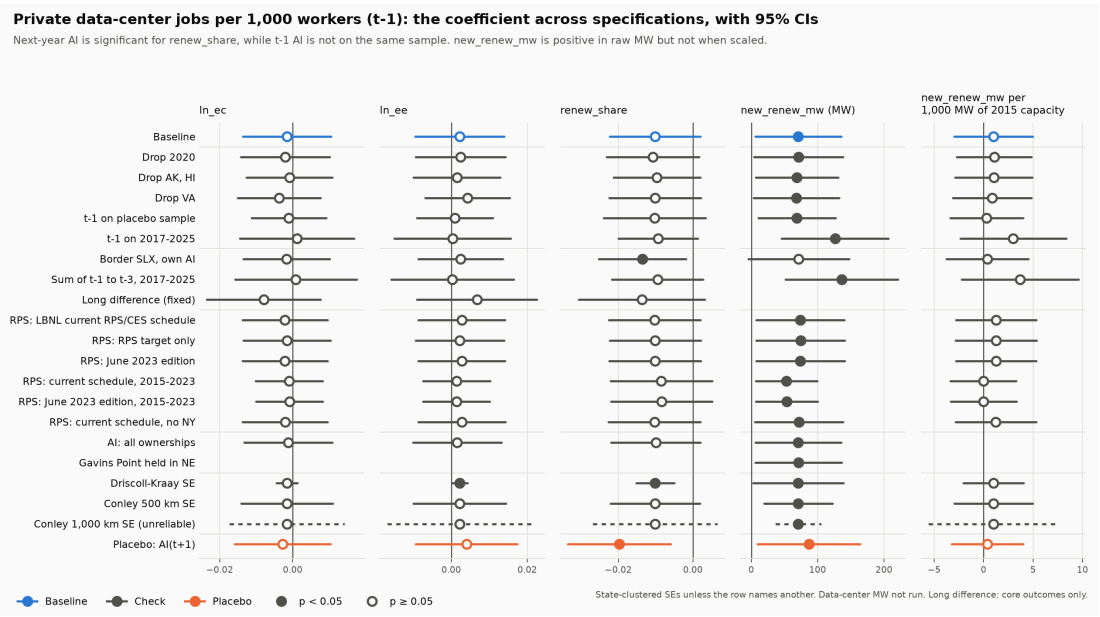


Figure 7. The AI coefficient across specifications. Next-year AI is significant for renew_share, while AI(t-1) isn't on the same sample.

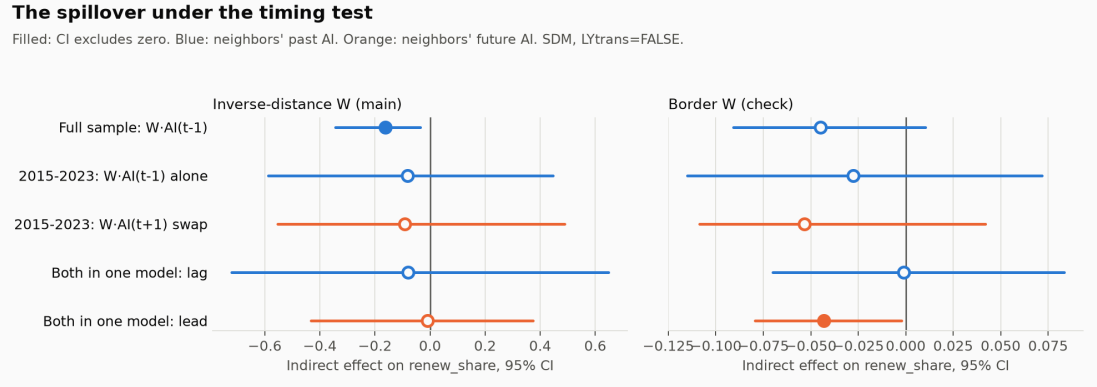


Figure 8. The renew_share spillover under the timing test. It isn't significant on 2015-2023, and with border weights neighbors' future AI carries the association.

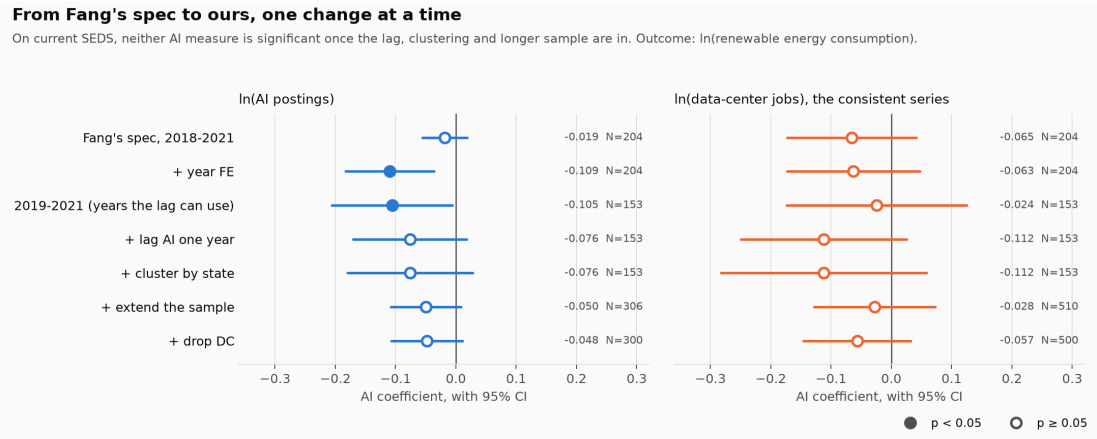


Figure 9. The AI coefficient from Fang's spec to ours. Data-center jobs are the more reliable extension measure.

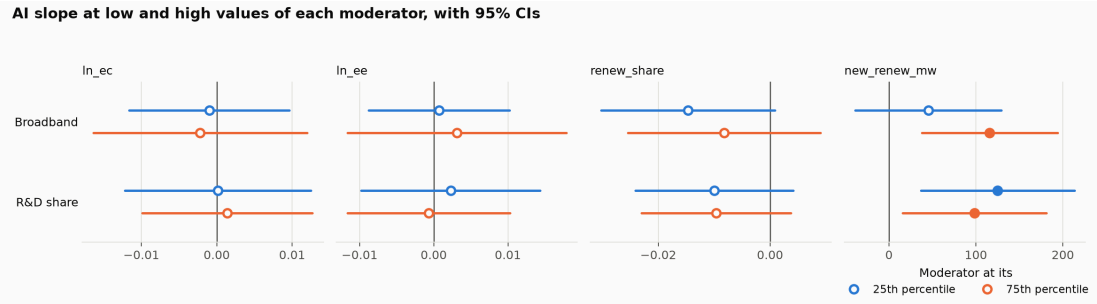


Figure 10. AI slope at the 25th and 75th percentiles of broadband and R&D share.

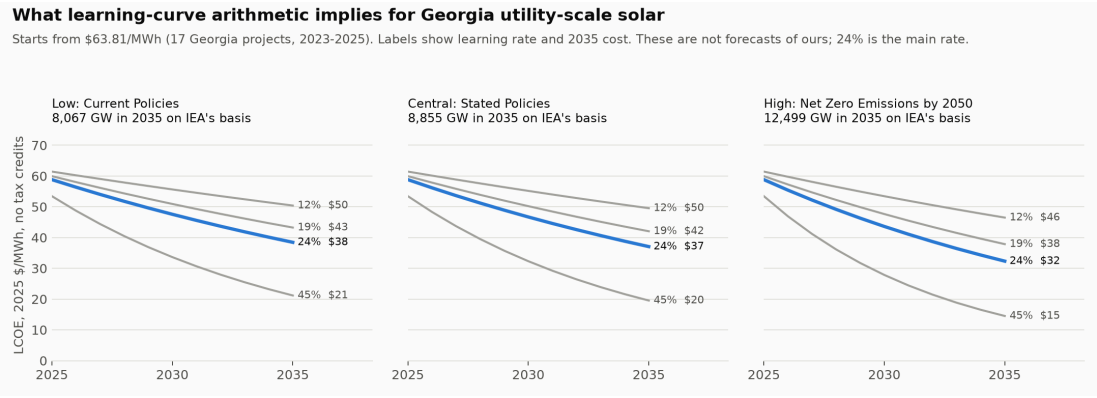


Figure 11. Georgia solar: learning-curve arithmetic to 2035 on IEA WEO 2025 paths (Annex A, Tables A.3a, A.3b and A.3c; pp. 417, 423 and 429). Not a forecast of ours.

8. What we can rule out

95% CI per one within-state SD of AI. **The SD used is 0.5212: winsorized lagged AI after removing both state and year effects, 50 states, N = 550.** That's the variation the Zhang models use. The Fang rows use the SD of each log AI measure the same way. A typical swing in data-center jobs isn't associated with a fall in the renewable share of more than 1.16 points. The capacity row's range excludes zero, but that result isn't robust to scaling and fails the timing test. These limits apply to job swings seen in this period, not to a much larger future build-out.

Table 21. Per one within-state SD of AI. Estimate and 95% CI.

Outcome	Test	SD used	Estimate per SD	95% CI per SD	Unit
Electricity use per person	Zhang	0.5212	-0.08	[-0.71, 0.54]	percent
GDP per MWh	Zhang	0.5212	0.11	[-0.50, 0.73]	percent
Renewable share of generation	Zhang	0.5212	-0.53	[-1.16, 0.11]	percentage points
Change in renewable capacity	Zhang	0.5212	36.92	[3.05, 70.79]	MW
Same, per 1,000 MW of 2015 capacity	Zhang	0.5212	0.52	[-1.55, 2.59]	MW per 1,000 MW
Renewable consumption, per SD of ln(AI postings)	Fang	0.2457	-1.18	[-2.62, 0.27]	percent
Renewable consumption, per SD of ln(data-center jobs)	Fang	0.2163	-1.23	[-3.16, 0.70]	percent

9. Limitations

- **The AI measure.** Private NAICS 518210 jobs are a proxy for data-center activity, not AI use, and include general data processing and hosting. Jobs aren't MW. Georgia ranks above Virginia, likely from colocation and cloud hosting. BLS suppressed AK, ND, SD and WY in 2025.
- **Postings.** Postings aren't jobs. The vendor and counting changed across editions, and only 2019-2021 come from one edition.
- **Generation.** Utility-scale only, so no rooftop solar. Capacity is nameplate, and new capacity is a net change.
- **SEDS.** Ends in 2024. Renewable consumption includes non-electric uses, and EIA's 2023 accounting change (eia.gov/state/seds/seds-change) affects the Fang outcome.
- **RPS.** LBNL's current schedule, not the targets in force; LBNL revised past years.
- **Inference.** Little within-state AI variation over 11 years. Residual spatial dependence after two-way FE means clustered SEs may be too small. There's no strong instrument, so nothing here supports a causal reading.
- **Georgia.** Filings are the utility's own numbers, with project names redacted. The Q1-Q4 2024 large-load reports are pending: the PSC's list returns 404 to scripts. Georgia's regulated setting may not generalize.
- **Weather.** NOAA degree days cover the 48 contiguous states and enter one check only. Fang's extreme-weather split uses our count of NOAA Storm Events heat and cold records, since Fang's rule isn't stated.
- **Extensions.** No innovation dimension in the AI index; talent rests on two components; AI investment by state covers one year; NREL resource data weren't reachable, so capacity factors stand in.

10. What's next

- **Team review.** Read the readings and flags, by the Nov 17 core-results date.
- **By hand:** the Q1-Q4 2024 large-load reports (PSC Docket 55378), LBNL Queued Up (for the queue outcome), then rerun the affected steps.
- **Draft by Nov 24.** Start from `docs/2026-11-07/methods_section.tex`, which hasn't been compiled here, and `results/2026-11-07/paper/findings.md`. Turn-in is Dec 1.
- **GitHub.** Review `release/github/CHECKLIST.md` before pushing the clean copy. Don't push the working repo.

Appendix: where every number comes from

Built by `code/python/04k_results_report.py` with no new estimation. Tables are verbatim copies of `results/2026-11-07/paper/tables/`, built by `code/python/04g_results_pack.py` from the result files below, and each result row carries a parseable tag. Numbers in the text are read from result files as the report is built, then re-read and checked (135 of 135 match; 8 published figures are cited with their location). Log: `results/2026-11-07/paper/number_check_report.csv`.

A. Tables

Table	Table file	Result files
Table 1	results/2026-11-07/paper/tables/table1_descriptives.csv	data/out/state_panel_2015_2025.csv
Table 2	results/2026-11-07/paper/tables/table2_h1.csv	results/2026-11-02/h1/h1_core.csv; results/2026-11-02/h1/h1_extras.csv
Table 3	results/2026-11-07/paper/tables/table4_mediation.csv	results/2026-11-03/h2/h2_results.csv
Table 4	results/2026-11-07/paper/tables/table5b_sdm.csv	results/2026-11-03/h3_mainspec/sdm_core.csv; sdm_impacts.csv
Table 5	results/2026-11-07/paper/tables/table5c_spillover_placebo.csv	results/2026-11-04/h3_placebo/placebo_slx.csv; placebo_sdm_core.csv; placebo_sdm_impacts.csv
Table 6	results/2026-11-07/paper/tables/table6_robustness.csv	results/2026-11-04/robustness/robustness.csv
Table 7	results/2026-11-07/paper/tables/table7_fang_replication.csv	results/2026-11-04/fang/fang_results.csv
Table 8	results/2026-11-07/paper/tables/table8_fang_bridge.csv	results/2026-11-04/fang/fang_results.csv
Table 9	results/2026-11-07/paper/tables/table9_fang_moderation.csv	results/2026-11-04/fang/fang_results.csv
Table 10	results/2026-11-07/paper/tables/table10_fang_by_type.csv	results/2026-11-04/fang/fang_results.csv
Table 11	results/2026-11-07/paper/tables/table13_fang_placebo.csv	results/2026-11-04/fang/fang_results.csv
Table 12	results/2026-11-07/paper/tables/table22_fang_table2_by_basis.csv	results/2026-11-07/fang/fang8_results.csv
Table 13	results/2026-11-07/paper/tables/table23_gap_decomposition.csv	results/2026-11-07/fang/fang8_results.csv
Table 14	results/2026-11-07/paper/tables/table33_ext_drivers_ai.csv	results/2026-11-07/extend/s4_drivers_models.csv
Table 15	results/2026-11-07/paper/tables/table36_ext_slope_ranks.csv	results/2026-11-07/extend/p2b_permutation_ranks.csv
Table 16	results/2026-11-07/paper/tables/table39_ext_index_family_summary.csv	results/2026-11-07/extend/p2b_index_family.csv
Table 17	results/2026-11-07/paper/tables/table15_iv_first_stage.csv	results/2026-11-04/stretch/stretch.csv
Table 18	results/2026-11-07/paper/tables/table16_iv.csv	results/2026-11-04/stretch/stretch.csv
Table 19	results/2026-11-07/paper/tables/table17_moderation.csv	results/2026-11-04/stretch/stretch.csv
Table 20	results/2026-11-07/paper/tables/table18_georgia.csv	results/2026-11-04/georgia_solar/projection_long.csv
Table 21	results/2026-11-07/paper/tables/table19_rule_out.csv	results/2026-11-07/paper/rule_out.csv

B. Numbers in the text

Number	File (under results/ unless shown)	Result tag or location	Column
2.3	our approximation, EXPERIMENTS.md F18	published	
2.9	our approximation, EXPERIMENTS.md F18	published	
-0.0016	2026-11-02/h1/h1_core.csv	h1_twfe_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_50s2015_2025	estimate
0.795	2026-11-02/h1/h1_core.csv	h1_twfe_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_50s2015_2025	pval
0.0022	2026-11-02/h1/h1_core.csv	h1_twfe_ai518210_lag1_ctrl4_w9901_cl-state_ln_ee_50s2015_2025	estimate
0.707	2026-11-02/h1/h1_core.csv	h1_twfe_ai518210_lag1_ctrl4_w9901_cl-state_ln_ee_50s2015_2025	pval
-0.0101	2026-11-02/h1/h1_core.csv	h1_twfe_ai518210_lag1_ctrl4_w9901_cl-state_renew_share_50s2015_2025	estimate

Number	File (under results/ unless shown)	Result tag or location	Column
0.100	2026-11-02/h1/h1_core.csv	h1_twfe_ai518210_lag1_ctrl4_w9901_cl-state_renew_share_50s2015_2025	pval
-0.0210	2026-11-04/robustness/robustness.csv	rob_placebo-joint-lead_ai518210_lead1lag1_ctrl4_w9901_cl-state_renew_share_50s2015_2024	estimate
0.007	2026-11-04/robustness/robustness.csv	rob_placebo-joint-lead_ai518210_lead1lag1_ctrl4_w9901_cl-state_renew_share_50s2015_2024	pval
0.0015	2026-11-04/robustness/robustness.csv	rob_placebo-joint-lag_ai518210_lead1lag1_ctrl4_w9901_cl-state_renew_share_50s2015_2024	estimate
0.827	2026-11-04/robustness/robustness.csv	rob_placebo-joint-lag_ai518210_lead1lag1_ctrl4_w9901_cl-state_renew_share_50s2015_2024	pval
70.8	2026-11-02/h1/h1_extras.csv	h1x_twfe_ai518210_lag1_ctrl4_w9901_cl-state_new_renew_mw_50s2015_2025	estimate
0.033	2026-11-02/h1/h1_extras.csv	h1x_twfe_ai518210_lag1_ctrl4_w9901_cl-state_new_renew_mw_50s2015_2025	pval
1.00	2026-11-02/h1/h1_extras.csv	h1x_rob_twfe_ai518210_lag1_ctrl4_w9901_cl-state_new_renew_per1000mw_50s2015_2025	estimate
0.616	2026-11-02/h1/h1_extras.csv	h1x_rob_twfe_ai518210_lag1_ctrl4_w9901_cl-state_new_renew_per1000mw_50s2015_2025	pval
-0.000157	2026-11-03/h2/h2_results.csv	h2a_med_ai518210_lag1_M-techshare_ctrl4_w9901_cl-state_renew_share_50s2015_2025_eq-a	estimate
0.454	2026-11-03/h2/h2_results.csv	h2a_med_ai518210_lag1_M-techshare_ctrl4_w9901_cl-state_renew_share_50s2015_2025_eq-a	pval
0.322	2026-11-07/paper/tables/mediation_summary.csv	h2_sobel_summary	min_p
0.991	2026-11-07/paper/tables/mediation_summary.csv	h2_sobel_summary	max_p
-0.161	2026-11-03/h3_mainspec/sdm_impacts.csv	h3_sdm-ML_ai518210_lag1_ctrl4_w9901_ml_renew_share_50s2015_2025_invdist50	plugin
0.035	2026-11-03/h3_mainspec/sdm_impacts.csv	h3_sdm-ML_ai518210_lag1_ctrl4_w9901_ml_renew_share_50s2015_2025_invdist50	p
0.0531	Fang et al. 2025, Table 2, col. 6	published	
-0.0188	2026-11-04/fang/fang_results.csv	fang_rep_c6_aipostings_contemp_state_iid_lnET_51s2018_2021_AI	estimate
0.317	2026-11-04/fang/fang_results.csv	fang_rep_c6_aipostings_contemp_state_iid_lnET_51s2018_2021_AI	pval
0.0348	2026-11-07/fang/fang8_results.csv	fang8_t2_c6-archived_aipostings_contemp_state_iid_lnET_arch_51s2018_2021_AI	estimate
0.057	2026-11-07/fang/fang8_results.csv	fang8_t2_c6-archived_aipostings_contemp_state_iid_lnET_arch_51s2018_2021_AI	pval
-0.0479	2026-11-04/fang/fang_results.csv	fang_bridge_s6-noDC_aipostings_lag1_state-year_cl-state_lnET_50s2019_2024_AI	estimate
0.109	2026-11-04/fang/fang_results.csv	fang_bridge_s6-noDC_aipostings_lag1_state-year_cl-state_lnET_50s2019_2024_AI	pval
-0.0567	2026-11-04/fang/fang_results.csv	fang_bridge_s6-noDC_dcjobs_lag1_state-year_cl-state_lnET_50s2015_2024_AI	estimate
0.208	2026-11-04/fang/fang_results.csv	fang_bridge_s6-noDC_dcjobs_lag1_state-year_cl-state_lnET_50s2015_2024_AI	pval
-0.0477	2026-11-07/extend/s2_bytype_family.csv	ext_bytype_twfe_aipostp1000_lag1_ctrl4_w9901_cl-state_ln_gen_wind_50s	estimate
-0.012	2026-11-07/extend/s4_drivers_models.csv	ext_natl_statefe_ai518210_lag1_ctrl4_w9901_cl-state_renew_share_50s	estimate
-0.011	2026-11-07/extend/s4_drivers_models.csv	ext_natl_drivers_ai518210_lag1_ctrl4_w9901_cl-state_renew_share_50s	estimate
-0.010	2026-11-07/extend/s4_drivers_models.csv	ext_natl_twfe_ai518210_lag1_ctrl4_w9901_cl-state_renew_share_50s	estimate
2.81	2026-11-04/stretch/stretch.csv	stretch_iv_base2015_ai518210_lag1_ctrl4_w9901_cl-state_first_stage_50s2015_2025_KP_F	estimate
37.06	2026-11-04/georgia_solar/projection_long.csv	ga_solar_lc_lr24_cap-central_2035	lcoe_2025usd_mwh

Number	File (under results/ unless shown)	Result tag or location	Column
14.51	2026-11-04/georgia_solar/projection_long.csv	ga_solar_lc_lr45_cap-high_2035	lcoe_2025usd_mwh
50.40	2026-11-04/georgia_solar/projection_long.csv	ga_solar_lc_lr12_cap-low_2035	lcoe_2025usd_mwh
0.5212	2026-11-07/paper/rule_out.csv	rob_base_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_50s2015_2025	sd_ai
-0.71	2026-11-07/paper/rule_out.csv	rob_base_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_50s2015_2025	ci_lo_per_sd
0.54	2026-11-07/paper/rule_out.csv	rob_base_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_50s2015_2025	ci_hi_per_sd
-0.50	2026-11-07/paper/rule_out.csv	rob_base_ai518210_lag1_ctrl4_w9901_cl-state_ln_ee_50s2015_2025	ci_lo_per_sd
0.73	2026-11-07/paper/rule_out.csv	rob_base_ai518210_lag1_ctrl4_w9901_cl-state_ln_ee_50s2015_2025	ci_hi_per_sd
-1.16	2026-11-07/paper/rule_out.csv	rob_base_ai518210_lag1_ctrl4_w9901_cl-state_renew_share_50s2015_2025	ci_lo_per_sd
0.11	2026-11-07/paper/rule_out.csv	rob_base_ai518210_lag1_ctrl4_w9901_cl-state_renew_share_50s2015_2025	ci_hi_per_sd
15,600	data/georgia_large_load_table2.csv	{'row': 'Commitments'}	q2_2026_mid2030s
3,600	data/georgia_large_load_table2.csv	{'row': 'Commitments'}	ifiling_mid2030s
550	2026-11-02/h1/h1_core.csv	h1_twfe_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_50s2015_2025	N
223	2026-11-07/paper/validation_counts.csv	validation_counts_all_total	n
178	2026-11-07/paper/validation_counts.csv	validation_counts_all_green	n
43	2026-11-07/paper/validation_counts.csv	validation_counts_all_yellow	n
1	2026-11-07/paper/validation_counts.csv	validation_counts_all_red	n
1	2026-11-07/paper/validation_counts.csv	validation_counts_all_notrun	n
20	2026-11-07/paper/validation_counts.csv	validation_counts_model_variables_total	n
17	2026-11-07/paper/validation_counts.csv	validation_counts_model_variables_green	n
3	2026-11-07/paper/validation_counts.csv	validation_counts_model_variables_yellow	n
0	2026-11-07/paper/validation_counts.csv	validation_counts_model_variables_red	n
0.001	2026-11-04/robustness/robustness.csv	rob_dk_ai518210_lag1_ctrl4_w9901_dk_renew_share_50s2015_2025	pval
0.094	2026-11-03/h3/h3_moran.csv	{'variable': 'renew_share', 'W_version': 'invdist50', 'type': 'raw', 'year': 2015}	moran_I
0.157	2026-11-03/h3/h3_moran.csv	{'variable': 'renew_share', 'W_version': 'invdist50', 'type': 'raw', 'year': 2025}	moran_I
0.0419	2026-11-04/h3_placebo/placebo_slx.csv	h3_slx-placebo-d_both_ai518210_lead1lag1_ctrl4_w9901_cl-state_ln_ec_50s2015_2023_invdist50_theta_lag	estimate
0.034	2026-11-04/h3_placebo/placebo_slx.csv	h3_slx-placebo-d_both_ai518210_lead1lag1_ctrl4_w9901_cl-state_ln_ec_50s2015_2023_invdist50_theta_lag	pval
0.0234	2026-11-04/h3_placebo/placebo_slx.csv	h3_slx-placebo-d_both_ai518210_lead1lag1_ctrl4_w9901_cl-state_ln_ec_48s2015_2023_border48_theta_lag	estimate
0.024	2026-11-04/h3_placebo/placebo_slx.csv	h3_slx-placebo-d_both_ai518210_lead1lag1_ctrl4_w9901_cl-state_ln_ec_48s2015_2023_border48_theta_lag	pval
-0.0209	2026-11-04/h3_placebo/placebo_slx.csv	h3_slx-placebo-d_both_ai518210_lead1lag1_ctrl4_w9901_cl-state_ln_ee_48s2015_2023_border48_theta_lag	estimate
0.033	2026-11-04/h3_placebo/placebo_slx.csv	h3_slx-placebo-d_both_ai518210_lead1lag1_ctrl4_w9901_cl-state_ln_ee_48s2015_2023_border48_theta_lag	pval
-0.0197	2026-11-04/robustness/robustness.csv	rob_placebo_ai518210_lead1_ctrl4_w9901_cl-state_renew_share_50s2015_2024	estimate
0.006	2026-11-04/robustness/robustness.csv	rob_placebo_ai518210_lead1_ctrl4_w9901_cl-state_renew_share_50s2015_2024	pval
86.8	2026-11-04/robustness/robustness.csv	rob_placebo_ai518210_lead1_ctrl4_w9901_cl-state_new_renew_mw_50s2015_2024	estimate
0.028	2026-11-04/robustness/robustness.csv	rob_placebo_ai518210_lead1_ctrl4_w9901_cl-state_new_renew_mw_50s2015_2024	pval

Number	File (under results/ unless shown)	Result tag or location	Column
55.9	2026-11-04/robustness/robustness.csv	rob_placebo-joint-lead_ai518210_lead1lag1_ctrl4_w9901_cl-state_new_renew_mw_50s2015_2024	estimate
0.219	2026-11-04/robustness/robustness.csv	rob_placebo-joint-lead_ai518210_lead1lag1_ctrl4_w9901_cl-state_new_renew_mw_50s2015_2024	pval
37.5	2026-11-04/robustness/robustness.csv	rob_placebo-joint-lag_ai518210_lead1lag1_ctrl4_w9901_cl-state_new_renew_mw_50s2015_2024	estimate
0.191	2026-11-04/robustness/robustness.csv	rob_placebo-joint-lag_ai518210_lead1lag1_ctrl4_w9901_cl-state_new_renew_mw_50s2015_2024	pval
-0.0021	2026-11-04/robustness/weather_check.csv	rob_weather-hddcdd_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_48s2015_2025_ai_dc_per1000_lag1	estimate
0.711	2026-11-04/robustness/weather_check.csv	rob_weather-hddcdd_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_48s2015_2025_ai_dc_per1000_lag1	pval
-0.0009	2026-11-04/robustness/weather_check.csv	rob_weather-base48_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_48s2015_2025_ai_dc_per1000_lag1	estimate
0.880	2026-11-04/robustness/weather_check.csv	rob_weather-base48_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_48s2015_2025_ai_dc_per1000_lag1	pval
0.001	2026-11-04/robustness/weather_check.csv	rob_weather-hddcdd_ai518210_lag1_ctrl4_w9901_cl-state_ln_ec_48s2015_2025_hdd_k	pval
3,412	EIA SEDS change notice	published	
0.0346	2026-11-07/fang/fang8_results.csv	fang8_t2_c1-archived_aipostings_contemp_state_iid_lnET_arch_51s2018_2021_AI	estimate
0.0435	2026-11-07/fang/fang8_results.csv	fang8_t2_c2-archived_aipostings_contemp_state_iid_lnET_arch_51s2018_2021_AI	estimate
0.0137	2026-11-07/fang/fang8_results.csv	fang8_t2_c6-rebuilt_aipostings_contemp_state_iid_lnET_rb_51s2018_2021_AI	estimate
0.500	2026-11-07/fang/fang8_results.csv	fang8_t2_c6-rebuilt_aipostings_contemp_state_iid_lnET_rb_51s2018_2021_AI	pval
61%	2026-11-07/extend/tables/p2_gap_decomposition.csv	{'ai_measure': 'ln AI postings', 'column': 6}	share_accounting_rebuild
11.79	2026-11-04/fang/seds_basis_level_check.csv	fang_level_1	mean_lnET
11.78	2026-11-04/fang/seds_basis_level_check.csv	fang_level_4	mean_lnET
11.90	2026-11-04/fang/seds_basis_level_check.csv	fang_level_6	mean_lnET
11.44	2026-11-04/fang/seds_basis_level_check.csv	fang_level_3	mean_lnET
-0.0419	2026-11-07/extend/p2_checks.csv	ext_p2_p1trend_fang6trend_aipostings_lnET_arch_51s2018_2021_AI	estimate
0.121	2026-11-07/extend/p2_checks.csv	ext_p2_p1trend_fang6trend_aipostings_lnET_arch_51s2018_2021_AI	pval
0.2980	2026-11-07/extend/p2_checks.csv	ext_p2_p1placebo_fang6_petcare_lnET_arch_51s2018_2021_AI	estimate
0.003	2026-11-07/extend/p2_checks.csv	ext_p2_p1placebo_fang6_petcare_lnET_arch_51s2018_2021_AI	pval
0.708	2026-11-07/extend/e1_descriptives_vs_tableA1.csv	ext_a1_lnEPU_	diff
0.014	2026-11-07/fang/fang8_results.csv	fang8_es_current_aishare2019_event_state-year_cl-state_lnET_50s2015_2024_joint_pre_2015_2020	pval
120	2026-11-04/fang/net_producer_check.csv	fang_netdef_gen_gt_sales_difference_aipostings_producer_minus_consumer	net_producer_state_years
40	2026-11-04/fang/net_producer_check.csv	fang_netdef_teprb_gt_tetcb_difference_aipostings_producer_minus_consumer	net_producer_state_years

Number	File (under results/ unless shown)	Result tag or location	Column
104	2026-11-04/fang/net_producer_check.csv	fang_netdef_elisb_negative_difference_aipostings_producer_minus_consumer	net_producer_state_years
0.0864	2026-11-04/fang/fang_results.csv	fang_inter_fang-net_producer_aipostings_contemp_state_iid_lnET_51s2018_2021_difference_high_minus_low	estimate
0.003	2026-11-04/fang/fang_results.csv	fang_inter_fang-net_producer_aipostings_contemp_state_iid_lnET_51s2018_2021_difference_high_minus_low	pval
0.316	2026-11-04/fang/net_producer_check.csv	fang_netdef_teprb_gt_tetcb_difference_aipostings_producer_minus_consumer	pval
0.177	2026-11-04/fang/net_producer_check.csv	fang_netdef_elisb_negative_difference_aipostings_producer_minus_consumer	pval
0.086	2026-11-07/extend/s2_bytype_family.csv	ext_bytype_twfe_aipostp1000_lag1_ctrl4_w9901_cl-state_ln_gen_wind_50s	qval
0.135	2026-11-07/extend/p2_checks.csv	ext_p2_p2bytype_twfe-trend_aipostp1000_ln_gen_wind_50s_AI	pval
0.076	2026-11-07/extend/s2_georgia_implied.csv	ext_bytype_twfe_ai518210_lag1_ctrl4_w9901_cl-state_share_solar_50s	actual_change
25%	2026-11-07/extend/s2_georgia_implied.csv	ext_bytype_twfe_ai518210_lag1_ctrl4_w9901_cl-state_share_solar_50s	at_most_share_of_actual
8,200	2025 IRP, p. 1 [PDF p. 8]	published	
9,500	2025 IRP, p. 3 [PDF p. 10]	published	
1,323	2025 IRP, p. 126 [PDF p. 133]	published	
-1.6%	2026-11-07/extend/tables/s3_bounds_per_within_sd.csv	ext_impact_twfe_ai518210_lag1_ctrl4_w9901_cl-state_ln_sales_com_pc_50s	per_sd_lo
1.2%	2026-11-07/extend/tables/s3_bounds_per_within_sd.csv	ext_impact_twfe_ai518210_lag1_ctrl4_w9901_cl-state_ln_sales_com_pc_50s	per_sd_hi
0.95	2026-11-07/extend/s4_path_fit.csv	ext_natl_pathfit_ai518210_drivers_renew_share	correlation
0.764	2026-11-07/extend/p2b_relevance_index.csv	ext_p2b_relevance_ai518210_ln_sales_com_pc_all50_pooled	pval
3	2026-11-07/extend/p2b_permutation_ranks.csv	ext_p2b_permrank_ai518210_all50_VA	rank_desc
7	2026-11-07/extend/p2b_permutation_ranks.csv	ext_p2b_permrank_ai518210_all50_TX	rank_desc
0.73	logs/2026-11-07/ai_index_diagnostics.csv	{'statistic': 'Cronbach alpha, 5 components'}	value
0.55	logs/2026-11-07/ai_index_crossstate_checks.csv	{'check': 'BTOS 2024 average share of firms using AI'}	pearson_index_a_2024
0.715	2026-11-07/extend/p2b_relevance_index.csv	ext_p2b_relevance_indexA_ln_sales_com_pc_all50_pooled	pval
0.014	2026-11-07/extend/p2b_distillery_rerun.csv	ext_p2b_distill_twfe_trend_lnRE_never_suppressed_states	pval
63.81	2026-11-04/georgia_solar/base_cost.csv	{'area': 'Georgia'}	lcoe_capacity_weighted
17	2026-11-04/georgia_solar/base_cost.csv	{'area': 'Georgia'}	n_projects
49.0	2026-11-04/georgia_solar/national_lcoe_lbnl_tab.csv	{'year': 2022}	lcoe
61.0	2026-11-04/georgia_solar/national_lcoe_lbnl_tab.csv	{'year': 2025}	lcoe
4,000	Georgia Power 2025 IRP, Main Document, p. 60 [PDF p. 67]	published	
550	2026-11-02/h1/h1_core.csv	h1_twfe_ai518210_lag1_ctrl4_w9901_cl-state_ln_ee_50s2015_2025	N

Number	File (under results/ unless shown)	Result tag or location	Column
1.16	2026-11-07/paper/rule_out.csv	rob_base_ai518210_lag1_ctrl4_w9901_cl-state_renew_share_50s2015_2025	ci_lo_per_sd
450	2026-11-04/h3_placebo/placebo_sdm_impacts.csv	h3_sdm-ML-placebo-a_lag_ai518210_lead1lag1_ctrl4_w9901_ml_renew_share_50s2015_2023_invdist50	N